

Program : Diploma in Wood and Paper Technology	
Course Code : 5122	Course Title: Wood Products-II
Semester : 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide basic knowledge regarding the manufacturing process of Wood based panel products such as Block board, particle board, fibre board, oriented strand board. etc,
- To make aware of the details of their properties and testing.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Wood processing		Wood processing	3
Processing of wood		Wood Products-I	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Outline the Manufacture of block boards, flush door, testing and properties.	15	Applying
CO2	Disseminate pressing, finishing, testing and properties of particle board	14	Applying
CO3	Explain the Chipping of wood, pulping methods, Mat formation manufacture of Insulation board and Hardboard	15	Applying
CO4	Explain sandwich composition, bulking treatments and testing and properties of wood laminates	14	Applying
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	1			2			
CO2	3				2		
CO3				3			
CO4	3			2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the Manufacture of block boards, flush door, testing and properties		
M1.01	Outline manufacturing process of block board	3	Remembering
M1.02	Explain exterior grade, interior grade, single layer, three layer, multi-layer, graded, extrusion pressed and flat presses P.B	7	Remembering
M1.03	Explain the schematic outline for P.B manufacture	3	Understanding
M1.04	Outline Need of Drying of PB	2	Applying
Contents: Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types - solid core flush door, cellular core (cellulose) flush door. Adhesive used. chipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blackboard-Particle board - introduction - history - particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B - exterior grade, interior grade, single layer, three layer, multi-layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials, adhesive - UF and PF resin, additives - water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer.			
CO2	Disseminate pressing, finishing, testing and properties.		
M2.01	Outline Chip storage and Drying of Chip	2	Understanding

M2.02	Summaries general considerations of chip blending	5	Applying
M2.03	Explain particle board	2	Understanding
M2.04	Enumerate the properties and testing of particle board	5	Remembering
	Series Test - I	1	

Contents:

Storage of chips - Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending - Definition, general consideration, batch operation, continuous operation, Mat laying - Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat- Pre-pressing and pressing of P.B - pre pressing methods - advantages and its disadvantages. Pressing - two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process - continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calendar Particleboard - Mande - Baisen System - P.B finishing - Sizing sanding or thickening by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B - Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and resistance against destruction. P.B testing methods - Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B.

CO3	Explain the Chipping of wood, pulping methods, mat formation, manufacture of Insulation board and Hardboard		
M3.01	Recall Fiberboard, raw materials for fiberboard, manufacturing process of fiberboard	6	Understanding
M3.02	Explain pulping processes	3	Applying
M3.03	Describe mat formation	3	Applying
M3.04	Develop testing of hardboard and MDF	3	Applying

Contents:

Fiber board insulation board MDF and hard board - history and development of fiber board - classifications of fiber board - modified wood raw materials - non wood fibrous raw materials - properties and uses of different types of fiber board - properties of semi rigid insulation board, rigid insulation board, hard boards - schematic outline of fiber board manufacturer using wet, semi dry and dry process - chipping - magnetic separation and

screening of fibers - storage of chips and the importance of the M.C of chips during storage - wetting of chips - moisture content of chip before pulping - pulping processes - ground wood pulping by pocket grinders - thermo mechanical process- mat formation - importance of mat formation - methods of mat formation - mat formation by deckle box - Fourdrinier method insulation board - sizing of insulation stock - mat formation - drying - trimming and finishing - Fibreboard - block diagram of hardboard manufactures - sizing - mat formation and Pressing - oil tempering, heat treatment, Humidification - testing methods of fiber board.

CO4	Explain sandwich composition, bulking treatments and testing and properties of wood laminates.		
M4.01	Oriented strand board	5	Remembering
M4.02	Outline Densified wood	4	Applying
M4.03	Explain sandwich composition	5	Applying
	Series Test - II	1	

Contents:

Oriented Strand board - Manufacturing Methods - wood wool board - Method of manufacture of wood wool board - Properties and uses - Cement Bonded Particleboard - Method of Manufacture - Moulded particle Board - Werzalit Process - Densified Wood - method of Preparation of raw materials - densification of Wood by Impregnation and Compression - Metalized Wood - Compreg - Dimensional stabilisation of wood - the dimensional stabilization such as cross lamination, external and Internal coating, reduction in hygroscopicity, heat stabilisation & cross linking-Define sandwich Composition - Explain the raw materials - Edge banding of Particle Board - Laminated PB - Bulking treatment - bulking treatments - Bulking treatments such as salt treatment, Sugar treatment, poly ethylene glycol treatment, wax treatment, Phenolic resin treatment - Properties and uses of impreg - acetylated wood - Testing and Properties.

Text / Reference:

T/R	Book Title/Author
T1	Franz. F.P Kollmann, Principles of wood science and technology Vol. II - Springer
T2	Thomas M Malon, Modern particle board and drying process of fiber board manufacturing

Online resources:

Sl. No	Website Link
1	https://www.researchgate.net/publication/265168265_Timber_and_wood_products
2	http://www.kstr.lth.se/fileadmin/kstr/pdf_files/Timber_Engineering_2017/Timber_and_engineered_wood_products.pdf
3	http://www.dec.ny.gov/docs/lands_forests_pdf/woodproducts.pdf
4	https://www.fpl.fs.fed.us/documnts/fplgtr/fpl_gtr190.pdf
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